NISHAL STANISLAUS SILVA, *PhD*

Computer Vision & ML Engineer | Industrial Inspection, IoT & Edge Deployment

Canadian Permanent Resident, Eligible to work in Canada without sponsorship

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Re: Senior / Staff Perception Engineer, Autonomy & Algorithms - Waabi

Dear Hiring Manager,

I am applying for the Senior / Staff Perception Engineer role at Waabi, at the Senior level. Most of my career has been perception work: turning camera and multi-sensor streams into decisions that hold up on deployed hardware. At MAS Holdings I shipped production computer vision for manufacturing, including the camera-to-print geometry alignment behind Promptly, now a commercial on-demand printing product with facilities in four countries. In my PhD and postdoc at the University of Trento I took deep learning detectors for streaming sensor data from prototype to device, reaching 14 ms inference on a Raspberry Pi 4 at F1 0.76, 74x faster than the DTW baseline (JAES 2026).

What draws me to Waabi is that a Toronto-based team is putting research into commercial service: customer operations with Volvo in Texas began this month, and the same end-to-end model is meant to serve both trucks and robotaxis. Waabi World, the company's closed-loop simulator, ties real-world data, simulation and rigorous evaluation together, which is how I already work.

Three parts of the role match what I have done:

- **Multimodal sensor fusion.** At McGill I fused an IMU gesture detector (LSTM) with audio pattern detection through a hierarchical state machine, F1 0.78 on 500 events (IEEE IS2 2025). On the EU MUSMET project I built a real-time pipeline that time-synchronizes audio, EEG and mixed-reality head/hand tracking from four performers at once.
- **Rigorous experimental validation.** I design benchmark suites and ablations before I trust a model (RNN vs. DTW, audio vs. MIDI, set sizes 1/3/10), published four open datasets with 9,800+ recordings, and recently ran a 100,000-trial benchmark of six scoring methods.
- **Production-quality code.** Python and C++17 are my primary languages; I ran C++/Python inference engines on embedded hardware, and introduced CI/CD and code review at MAS.

I have not yet driven projects on 3D detection or tracking of traffic agents, and I have not worked with LiDAR or self-driving data. What transfers directly is camera geometry and registration from Promptly, temporal state over noisy sensor streams, and a habit of proving a model on held-out real data before it ships. In my first months I would learn the domain through Waabi's datasets and simulator while contributing evaluation and fusion tooling to the perception stack.

I am a Canadian permanent resident based in Toronto, ON, eligible to work without sponsorship, and available immediately for onsite, hybrid or remote work. I would welcome the chance to discuss how my experience in real-time perception and sensor fusion can support the Autonomy & Algorithms team. My publications and open datasets are at nishal.xyz.

Sincerely,

Nishal Stanislaus Silva, PhD